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Service robots in healthcare: Toward a healthcare service robot acceptance model (sRAM)

Weng Marc Lim [a,b,c,d,*], K. Mohamed Jasim [e], A. Malathi [e]

[a] Sunway Business School, Sunway University, Sunway City, Selangor, Malaysia
[b] ASU-Cintana Alliance Global Partner Affiliate Faculty, Arizona State University, Tempe, AZ, USA
[c] School of Business, Law and Entrepreneurship, Swinburne University of Technology, Hawthorn, Victoria, Australia
[d] Global Research Centre, Sungkyunkwan University, Suwon, Gyeonggi, Republic of Korea
[e] VIT Business School, Vellore Institute of Technology, Vellore, Tamilnadu, India

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ABSTRACT

Our research examines the perceptions and intentions surrounding the use of healthcare service robots. Guided by service robot acceptance model (sRAM) and stimulus-organism-response (S-O-R) model, we explore how perceptions of functional, social, emotional, and robotic features of service robots shape their trust and use in healthcare. Our research incorporated data from 398 responses collected via an online questionnaire, which was analyzed using partial least squares structural equation modeling (PLS-SEM) through the SmartPLS software, revealing that functional (ease of use), emotional (anxiety and enjoyment), and social (social interactivity and presence) features significantly influence healthcare service robots trust and use. Contrarily, usefulness—a functional feature—had no significant role in shaping healthcare service robots trust and use. Nevertheless, trust mediated perceptions relating to anxiety, ease of use, enjoyment, social interactivity, and social presence with healthcare service robots use. Interestingly, anthropomorphism—a robotic feature—had no moderating effect while subjective norms—a non-robotic feature—only moderated the impact of social interactivity on healthcare service robots use. Conclusively, our research organizes sRAM antecedents into clear, discrete categories (functional, emotional, and social) and delivers a comprehensive, structured acceptance model. This new and novel model supports systematic theory development and comparability in healthcare service robot research while also offering critical implications for enhancing the integration and utilization of service robots within healthcare.

1. Introduction

1.1. Background of the research

Technology is integral to human existence, with innovations in artificial intelligence and sensory technology significantly enhancing service robots' functionality. These robots have transcended traditional industrial applications by venturing into human-centered areas and finding applications in domestic settings (Gao et al., 2025; Söderlund, 2022), educational institutions (Suhail et al., 2024; Wang et al., 2021), and servicescapes (Arikan et al., 2023; Byrd et al., 2024; Park et al., 2023), especially after COVID-19 (Merdin-Uygur & Ozturkcan, 2023), wherein contactless options emerge as a preference (Jiang et al., 2023; Li & Huang, 2022; Lim et al., 2024). Their application spans a wide array of industries, including but not limited to, medical (Cardona et al., 2025; Prentice & Nguyen, 2021), retail (De Gauquier et al., 2023; Vinoi et al., 2025), and travel (Liao & Huang, 2024; Pham et al., 2024) sectors, which, in turn, showcases their versatility and widespread utility.

Focusing on healthcare, technology implementations, including automation and service robots, contribute significantly to better quality, consistency, and effective service delivery (Irshad et al., 2025; Macalupu et al., 2025). Service robots have demonstrated their potential in assisting with different stages of patient care (Huang et al., 2023), catalyzing a surge in the medical service robots market, which is expected to reach a revenue of US$14 billion in 2025 and projected to increase at a 10 % annual growth rate through 2029 (Statista, 2025).

* Corresponding author. Sunway Business School, Sunway University, Sunway City, Selangor, Malaysia.
E-mail addresses: marcl@sunway.edu.my, lim@wengmarc.com, wengmarc@asu.edu, marclim@swin.edu.au (W.M. Lim), mohamedjasim.k@vit.ac.in (K.M. Jasim), malathi.a@vit.ac.in (A. Malathi).

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Yet, despite the said advancements, certain challenges remain. With an increasing aging population (Lim & Bowman, 2022), with one in six people in the world will be aged 60 years or over by 2030 (World Health Organization, 2024), and human capital inadequacies or pressures in healthcare (Lim, 2021; Yap et al., 2023), with the World Health Organization (2025) estimating a projected shortfall of 11 million healthcare workers by 2030, the need for automation and service robots in healthcare has never been more significant (Mishra & Pandey, 2023; Sahoo et al., 2023; Singh et al., 2025). This need is even more pronounced during global health crisis like COVID-19 (Merdin-Uygur & Ozturkcan, 2023), as service robots helped manage severe staff shortages in hospitals and reduced human contact by taking over tasks traditionally done by humans (Mamatha et al., 2024; Wang & Wang, 2021).

1.2. Practice-driven problem statement of the research

Service robots' importance in healthcare cannot be overstated. However, the readiness of patients to accept these service robots and their potential to enhance the patient experience are areas that remain relatively unexplored (Macalupu et al., 2025; Östlund et al., 2023). This gap in understanding represents a considerable challenge for healthcare providers looking to implement these technological advancements. While a great deal of research has investigated the adoption of artificial intelligence-supported technologies in sectors such as retail and services (Fuentes-Moraleda et al., 2020; Kao & Huang, 2023; Saputra et al., 2024) and its implications for leadership (Sposato, 2024; Zaidi et al., 2025), human resource management (Balcioğlu & Artar, 2024; Islam et al., 2023), and marketing (Thakur & Kushwaha, 2024), comparable studies focusing specifically on consumer perceptions and intentions toward service robots use in healthcare are notably lacking, which is problematic for several reasons.

Firstly, healthcare is a unique field with distinct dynamics that differentiate it from other sectors. Noteworthily, healthcare deals with highly personal and sensitive issues where the stakes are particularly high. Moreover, healthcare itself inherently evokes anxiety among consumers, who typically approach it as patients. Consequently, patient acceptance of robotic technology could be shaped by considerations other than usability alone.

Secondly, the lack of insight into patient perceptions and intentions may limit the successful integration of service robots into healthcare systems. Understanding their drivers is crucial in designing and implementing service robots that meet patient needs and expectations. This is especially true given that the effectiveness of these robots may be tied not only to their functional capabilities but also to their acceptance by patients, their families, and caregivers.

Thirdly, with the global healthcare crisis sparked by COVID-19 highlighting critical contributions that service robots can deliver, including managing staff shortages and reducing human contact, the need to understand and address patient perceptions and intentions regarding their use has become more critical than ever. Thus, investigating patients' perceptions and intentions toward using healthcare service robots is not only necessary but essential to unlock the full potential of this technology in a sector as vital as healthcare.

1.3. Theory-driven problem statement of the research

The importance of improving the theorization of models explaining the technological acceptance of service robots becomes evident when considering the complex landscape of factors influencing their adoption. While current models have heeded calls from technology acceptance proponents, such as Lim (2018), by surpassing the basic aspects of the models of technology acceptance and addressing the relationship between various factors, these efforts have yet to fully coalesce into a comprehensive understanding. Indeed, these factors influencing service robot acceptance remain largely fragmented and unorganized, a reality that becomes clear when we observe the prevailing lack of categorization. In particular, factors are seldom organized into meaningful categories, despite the establishment of rich multifaceted relationships among them (e.g., Fuentes-Moraleda et al., 2020; Kao & Huang, 2023). This fragmentation potentially hampers the potential of acquiring a thorough understanding of the dynamics at play and may hinder future development of effective strategies to promote service robot adoption. More importantly, this gap in theorization holds several significant implications.

Firstly, without a cohesive and organized model, it can be challenging to develop effective policies and schemes to promote service robot acceptance. This lack of organization hinders our ability to accurately comprehend and predict the potential acceptance of or resistance to these technologies.

Secondly, an organized model that categorizes factors into meaningful groups can deliver a finer-grained and organized understanding of the interplay between the factors influencing service robot acceptance. An organized categorization of factors would also provide a solid foundation for future research to build upon, guiding researchers toward areas that need further investigation and facilitating the generation of more precise hypotheses.

Thirdly, the healthcare sector is characterized by highly sensitive and deeply personal patient interactions, which, in turn, makes a fragmented understanding of acceptance factors particularly problematic. This incomplete perspective could lead researchers and practitioners to overlook, for instance, critical emotional and social dimensions that shape patients' trust and willingness to adopt service robots. Given that patients often experience heightened anxiety and vulnerability, neglecting these factors might result in robot designs and implementations that fail to adequately address patient concerns or meet their expectations. Consequently, a more comprehensive and systematically organized understanding is essential to ensure service robots effectively resonate with patients, caregivers, and families within healthcare settings.

1.4. Direction and scope of the research

The impetus to address the identified practical and theoretical gaps is not only significant but also imperative for the optimal utilization of service robots, especially in sectors as crucial as healthcare. Developing a more cohesive and well-structured service robot acceptance model (sRAM) is essential and would represent a considerable advancement in this space. Thus, our research's overarching goal is to create a comprehensive acceptance model specifically tailored for healthcare service robots.

Our envisioned model is comprehensive in several ways: *firstly*, it encapsulates diverse factors shaping service robots acceptance; *secondly*, it organizes these factors into coherent, meaningful categories; and *thirdly*, it posits clear hypotheses regarding the relationships among these factors, including those relating to direct (main), indirect (mediation), and interaction (moderation) effects. These relationships will be subjected to rigorous testing, with our research rooted in empirical evidence to ensure validity.

Our approach leverages two influential theoretical lenses to guide the development of the research model: Davis's (1989) information technology theory—i.e., the technology acceptance model (TAM)—and Mehrabian and Russell's (1974) environmental psychology theoretical framework—i.e., the stimulus-organism-response (S-O-R) model. The TAM grants a fundamental understanding of the manner in which technologies are accepted and serves as a foundational base for theoretical extension (Lim, 2018). The S-O-R model, in contrast, serves as an overarching, guiding structure for theorizing the relationships among an extended set of factors in this theory development process (Lim, Aggarwal, & Dandotiya, 2022).

Through integrating these theoretical frameworks, our research not only aims to explore the factors that, as a collective, presents a

comprehensive explanation of healthcare service robots acceptance, but it also endeavors to establish a rich, rigorous, yet parsimonious model. This model is designed to be thorough yet concise, providing detailed insights without becoming overburdened with complexity. In doing so, we strive to generate a better understanding of healthcare service robot acceptance that can guide future research and assist practitioners in this rapidly evolving field.

To this end, our research hopes to illuminate the path toward a more integrated and effective adoption of service robots in healthcare, paving the way for improved patient experiences, better care outcomes, and a more efficient healthcare delivery system. This endeavor aligns with the wider objectives of maximizing the potential benefits of emerging technologies in healthcare and contributing to the ongoing efforts to improve healthcare services' accessibility, effectiveness, and overall quality.

2. Literature review

2.1. Scope and role of service robots in healthcare

Robots, as technology-infused entities, possess characteristics that mimic human traits, bridging the gap between human abilities and technological advancements (Liu, Wang, & Wang, 2022; Premathilake & Li, 2024). The seminal work of Wirtz et al. (2018) conceptualize robots with customer-facing roles as independent, systemic platforms capable of engaging, conversing with, and providing services directly to the clientele of a firm. These machines first emerged in the medical field during the 1980s, initially designed to augment surgical procedures with robotic arm technology. Over time, artificial intelligence (AI) has enabled the expansion of these robots into other areas of healthcare (Wong et al., 2024).

According to the European Foresight Monitoring Network (EFMN), a healthcare robot is a system capable of executing coordinated mechatronic actions, facilitated by sensor technology, to support medical interventions, aid impaired individuals, and assist in the care, rehabilitation, and prevention programs for patients. These healthcare robots have since been utilized in a wide array of applications, including surgery, care, social assistance, walking aid, and rehabilitation (Kaur, 2024; Kyrarini et al., 2021).

Designed to provide assistance to humans, assistive robots enhance physical task execution and strengthen social engagement—they perform a variety of frontline services within hospitals, ranging from patient monitoring to assistance in daily activities (Barua, 2024), especially for older adults (Chua et al., 2025). Certain models even engage in direct conversational interaction with patients, providing cognitive stimulation and alerting nurses in cases of abnormal conditions (Gongor & Tutsoy, 2025).

Socially and physically assistive robots enhance direct, effective communication with humans. They have shown promise in improving cognitive abilities in patients with Alzheimer and dementia (Fan et al., 2025; Yuan et al., 2022), developing social- and work-related skills among people with autism (Doğan & Çolak, 2024; Yarrow et al., 2024), and facilitating emotional expression, verbalization, and socialization among patients at large (Foster et al., 2020; Rodríguez-Domínguez et al., 2024). Furthermore, rehabilitation robots offer assistance to individuals with physical impairments (Banyai & Brişan, 2024) while walking assistant robots cater to those with limited mobility (Sun et al., 2024).

Healthcare service robots are instrumental in streamlining routine tasks such as cleaning, bed preparation for patients, disinfecting hospital surfaces and handles, tracking medical inventories, assisting in laboratories, and more (McGinn et al., 2021; Jena et al., 2025). Moreover, AI-enabled robots equipped with medicine identifiers expedite the identification and distribution of medications to patients. Noteworthily, the introduction of healthcare service robots effectively reduces caregivers' workloads, thereby enabling more human interactions that enhance patients' emotional wellbeing. In addition, these robots have even been employed to fulfill receptionist duties within hospitals, further underscoring their versatility within healthcare settings.

2.2. From technology acceptance model (TAM) to service robot acceptance model (SRAM)

Previous research has often utilized TAM, as coined and popularized by Davis (1989), for discerning the factors driving individuals' technology use intention. This model positions usability factors in terms of ease of use alongside usefulness as core drivers of technology acceptance, serving as a foundational framework for building acceptance models across diverse domains (Lim, 2018; Oesterreich et al., 2025).

Over time, the TAM was expanded into its second and third versions by Venkatesh and Davis (2000) and Venkatesh and Bala (2008). These advancements integrated a broader array of determinants, including hedonic motivations, social influences, and the pivotal role of trust, thus enriching our repository of factors driving technology adoption. The evolution of TAM also led to the birth and recognition Venkatesh et al.'s (2003) theoretical integration in the field of information technology—as seen through the unified theory of acceptance and use of technology (UTAUT), a theory that redefined the core constructs of the original model to encompass effort and performance expectancies alongside the addition of facilitating conditions and social influence. Collectively, these frameworks construct a comprehensive theoretical foundation for dissecting the nuances of technology acceptance (Lim, 2018; Oesterreich et al., 2025).

Notwithstanding extant theoretical advances, service robots represent a distinct domain within technology acceptance. Service robots are essentially very different from alternative automated technologies as they perform a wide array of human activities and interact directly with humans (Mende et al., 2025; Söderlund, 2022). They are differentiated based on three main attributes: physical representation, human-like appearance, and task orientation (Lee et al., 2020; Lee & Yi, 2025; Wirtz et al., 2018). Hence, service robots need to serve both users' functional and socio-emotional or relational needs.

In light of these unique considerations, Wirtz et al. (2018) introduced a tailored framework known as the service robot acceptance model (sRAM), which amalgamates aspects from TAM and UTAUT with additional insights from Solomon et al.'s (1985) role theory alongside Fiske et al.'s (2002) stereotype content model. This framework is designed to explore the determinants of individuals' intentions to embrace service robots, suggesting that such decisions are shaped by a combination of functional characteristics (like ease of use and usefulness alongside norms) and socio-emotional elements (including the robots' perceived human-like qualities, interactive social capabilities, and presence), as well as relational dimensions (such as trust and rapport).

Although Wirtz et al.'s (2018) sRAM integrates socio-emotional and relational elements along with functional attributes to create a comprehensive framework for understanding service robot acceptance, subsequent developments of the sRAM have tended to deviate from the original categorization proposed by Wirtz and colleagues. Such a deviation, in turn, has resulted in the factors being presented in a somewhat fragmented and disorganized manner, as can be observed in studies such as those by Fuentes-Moraleda et al. (2020) and Kao and Huang (2023), where factors are presented and hypothesized independent of their categories. This issue extends into healthcare contexts as well, exemplified by Li et al. (2024), who explore anthropomorphic design attributes, perceived empathy, and interaction quality alongside traditional constructs such as effort and performance expectations, yet without organizing these factors according to clear conceptual categories.

Our position is that the categorization of factors is essential for several reasons. *Firstly*, categorization supports a structured, organized approach to understanding the diverse factors influencing service robot acceptance, facilitating a comprehensive analysis of patient trust and intention toward using service robots in healthcare. Secondly, within healthcare contexts, organizing constructs into clear categories ensures

that the full spectrum of patient-relevant considerations—functional, emotional, and social—is represented in the model, thereby preventing any inadvertent narrowing of its scope and the omission of critical drivers of patient acceptance. *Thirdly*, applying consistent categories across patient-focused studies promotes comparability and cumulative learning, enabling researchers and practitioners to systematically build knowledge on how different factors shape healthcare service robot uptake. Consequently, we advocate for a reemphasis on categorization in future adaptations of the sRAM to ensure this model continues to provide a comprehensive lens for investigating and understanding service robot acceptance in healthcare.

While the comprehensive scope of Wirtz et al.'s (2018) sRAM is duly appreciated, we also question some of its categorizations. Specifically, we find the classification of subjective social norms as a functional attribute to be misaligned given the inherently social nature of this factor. Furthermore, we find that merging social and emotional features into one category overlooks the differences between these two types of factors. Despite these disagreements, we concur on the fundamental importance of factor categorization in the model. Hence, we propose that refining and extending these categorizations could be instrumental in broadening our understanding of service robot acceptance, thereby contributing to knowledge advancement in this field.

2.3. Stimulus-organism-response (S-O-R) model for healthcare service robots

Developed and made popular by Mehrabian and Russell (1974), the S-O-R model is a widely accepted and robust theoretical lens for understanding the dynamics shaping service robots acceptance in healthcare. According this model, *stimuli* affect one's internal state, thereby leading to behavior change. In the context of our research, we considered a range of direct and moderating stimuli. The *direct stimuli* are perceptual and comprise *functional stimuli* like one's assessment of ease of use and usefulness, *emotional stimuli* involving one's supposed anxiety and enjoyment, and *social stimuli* such as social interactivity and presence. The *moderating stimuli* are evaluative and contain the degree of anthropomorphism attributed to the robot (i.e., as an example of a robot characteristic) and subjective norms relating to robot usage (i.e., as an example of a non-robot characteristic), which could moderate the impact of the direct, perceptual stimuli.

The *organism*, on the other hand, encapsulates the internal state in response to these stimuli. In the current investigation, this internal state is reflected through the individual's trust in the service robot. Noteworthily, trust represents the individual's psychological disposition or state toward service robots within the healthcare setting.

The *response* phase is the last stage and corresponds to healthcare service robots use. Usage intention is deemed to be a good proxy for actual use given that the former is a precursor of the latter, reflecting the individual's behavioral response to their trust in the service robot. Moreover, it echoes the individual's functional, emotional, and social state toward the service robot in tandem with the moderating role of stimuli such as anthropomorphism of and subjective norms toward healthcare service robots.

In applying the S-O-R model in our investigation, we present a model to effectively elucidate how functional, emotional, and social factors shape service robots use in healthcare settings. Examining these elements, we derive a more profound comprehension pertaining to individuals' trust formation in this technology, a factor that significantly encourages healthcare service robots use.

Building upon the aforementioned insights from the extant body of knowledge, Fig. 1 presents our proposed conceptual model. This model integrates the TAM and S-O-R model with core elements of functional, emotional, and social factors alongside their corresponding influence on trust and the moderating roles of robot and non-robot characteristics. Taken collectively, this allows us to depict a fuller picture of healthcare service robots acceptance. We believe this enriched model could offer

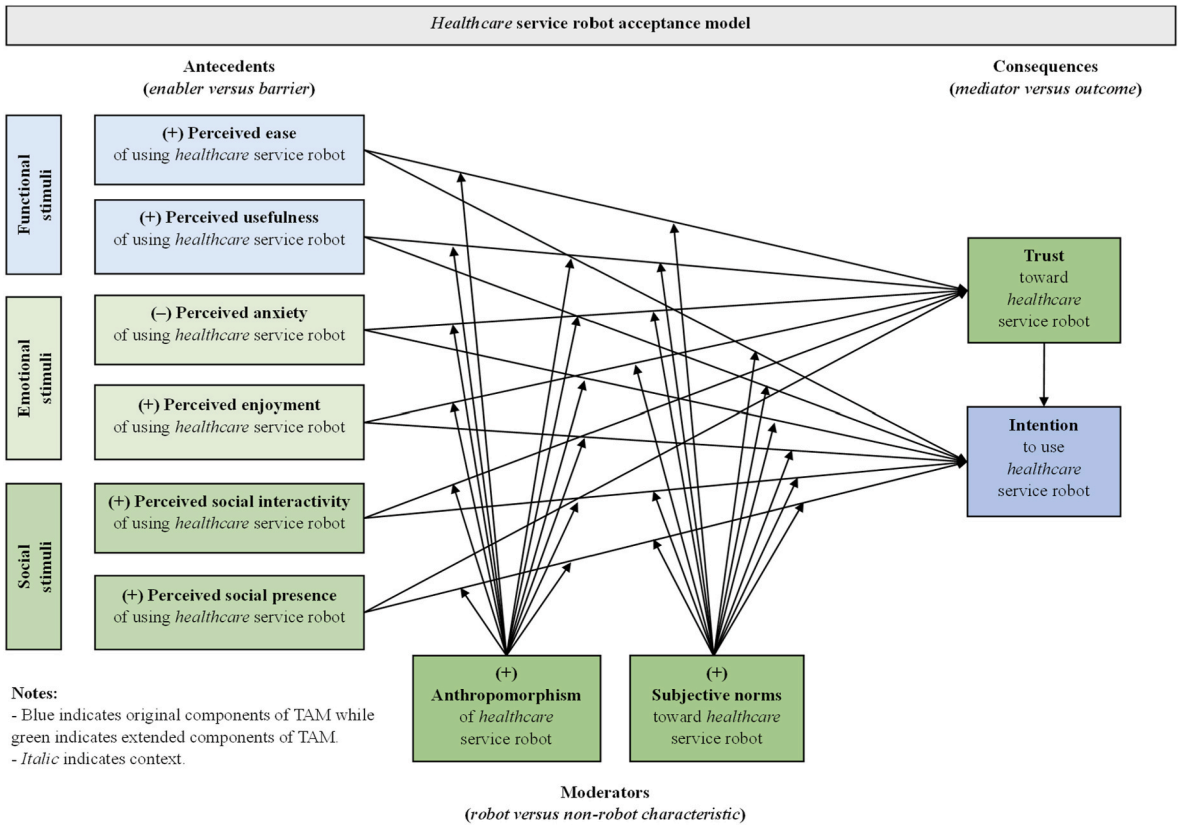


Fig. 1. Healthcare service robot acceptance model.

valuable insights to researchers and practitioners seeking to optimize service robots acceptance and use in healthcare.

3. Conceptual model and hypotheses development

3.1. The role of functional stimuli in healthcare service robots

The concept of robot acceptance is epitomized by the willing integration of healthcare robots into a person's life, and this integration is predicated upon three primary tenets: utility, convenience, and a lack of physical or emotional complexity in using these robots (Wirtz et al., 2018). Central to this acceptance are functional elements, a cornerstone of the TAM (Davis, 1989; Lim, 2018). As fundamental components of these functional elements, ease of use along with usefulness have both consistently demonstrated positive and significant impacts on individual behaviors. As Wirtz et al. (2018) argue, these functional elements have a robust correlation with consumer service robots acceptance; as ease of use and usefulness rise, so does individuals' acceptance of these robots.

3.1.1. The impact of perceived ease of use

Within the domain of healthcare service robots, perceived ease of use is pivotal, epitomized by the degree one believes that employing the technology will be effortless (Davis, 1989). Within this domain, it encapsulates the anticipation that interacting with a service robot will not demand excessive effort, a notion that has been linked to a heightened propensity for adopting these robotic aids (Pillai & Sivathanu, 2020). The interplay between this perceived ease and the development of trust in service robots has been scrutinized in various studies like Nangin et al. (2020) and Schnall et al. (2015), which extend to diverse areas including technology adoption and consumer repurchase intentions, respectively.

A wealth of research, including works by Park et al. (2018) and Schultz and Kumar (2024), has shed light on the nexus between perceptions of ease of use and the inclination to utilize given technologies. Notably, Bröhl et al. (2019) explored this relationship within the ambit of human-robot interactions across diverse cultural landscapes, revealing that an enhanced perception of ease of use fosters more favorable behavioral intentions. Moreover, investigations by scholars such as Seo and Lee (2021) have elucidated the indirect influence of ease of use on revisits in dining establishments that integrate service robots. Accordingly, we propose:

H1. Perceived ease of use has a significant impact on trust (H_{1a}) and intention to use healthcare service robots (H_{1b}).

3.1.2. The impact of perceived usefulness

The role of perceived usefulness stands as a cornerstone in technology acceptance, serving as a critical signal of an individual's intention to embrace new technologies. Perceived usefulness, as initially outlined by Davis (1989), relates to one's belief in the potential of a given system to improve their job performance. Within the domain of service robotics, the utility perceived by end-users significantly shapes their inclination toward adopting these technological aids.

Empirical evidence, drawing from studies by Lee et al. (2018) and further supported by Ventre and Kolbe (2020), underscores a direct linkage between the perceived utility of service robots and the trust users place in them. In particular, technologies that clearly demonstrate value by addressing users' needs or problems engender greater trust in the technology's efficacy (Chen & Zhu, 2024). This trust, in turn, is a vital precursor to the actual usage intentions (Tan et al., 2025). Noteworthily, the criticality of such intentions cannot be overstated, as they enable users to leverage the full spectrum of benefits offered by service robot technologies. The studies by Alshammari and Babu (2025) and Kim, Merrill, et al. (2024) highlight a positive interplay between perceived usefulness of technologies and intentions to utilize them. Echoing these findings, Lee (2025) show that the benefits of adopting service robots significantly shape the intention to adopt them, emphasizing how it shapes users' trust, which is attitudinal in nature (Nguyen, 2022), and, consequently, their behavioral intentions. Therefore, when users feel a technology is truly useful for them, they are not only more likely to find it trustworthy but are also far more inclined to embrace it in their lives. As a result, we propose:

H2. Perceived usefulness significantly influences trust (H_{2a}) and intention to use healthcare service robots (H_{2b}).

3.2. The role of emotional stimuli in healthcare service robots

The evaluation of individuals' behavior toward service robots suggests that innate interests and emotions play crucial roles. Informed by existing literature, we understand that these interests are sustained throughout individuals' interactions with service robots. Noteworthily, research has shown that interruptions to an individual's routine can trigger emotional responses (Baldessarelli, 2021), and hence, we can reasonably assumed service robots, being potential disruptors of routine, may elicit strong emotional reactions.

The basic psychological emotions—e.g., happiness, joy, surprise, fear, anger, sadness, and disgust—are often the initial emotional responses to any situation. Both positive and negative emotions can significantly influence subsequent cognitive and behavioral outcomes.

Human psychological reactions like anxiety and enjoyment toward service robots are shaped by the nature of the situation and the perceived social relationship between humans and robots. However, the emotions experienced by patients toward healthcare service robots can significantly differ from typical human-robot interactions, making emotion a pivotal driver influencing service robots acceptance in healthcare Lu et al., 2019). Therefore, emotional responses can strongly determine the acceptance or rejection of technology, thereby warranting attention in technology acceptance studies (Gursoy et al., 2019).

3.2.1. The impact of perceived anxiety

Perceived anxiety induced by robots is defined as the emotional state of anxiousness that prevents people from interacting with robots that possess communication capabilities in everyday life scenarios, particularly within the context of human-robot communication. Previous research confirms that anxiety significantly predicts usage intention and the quality of one's engagement with technology (Soren & Chakraborty, 2024). At times, anxiety induced by robots may deter individuals from engaging with these technologies, thereby affecting their usage intention (Huang et al., 2021). When a robot fails to meet a patient's needs, behaves unpredictably, or inadvertently harms the patient, negative emotions such as anxiety, fear, and mistrust may arise. Moreover, the degree of state anxiety experienced by a user before and during the interaction can greatly influence their trust in the robot (Miller et al., 2021). However, Lee et al. (2020), in their investigation of the likelihood of older adults to accept and use service robots, reveal that perceived anxiety did not substantially deter their adoption intentions. This outcome suggests that apprehensions associated with technology adoption may not uniformly affect everyone the same way. The findings prompt a deeper inquiry into the diverse factors influencing technology acceptance, underscoring the necessity for further research. Consequently, we propose:

H3. Perceived anxiety has a significant and negative impact on trust (H_{3a}) and the intention to use healthcare service robots (H_{3b}).

3.2.2. The impact of perceived enjoyment

Perceived enjoyment relates to the degree of pleasure or fun that can be obtained from engaging with a particular activity, including utilizing a technological solution, beyond the anticipated benefits or results (Davis et al., 1992). As an intrinsic motivator, perceived enjoyment significantly influences individuals' behavior (Venkatesh et al., 2002). Noteworthily, individuals can anticipate the outcomes of technological interactions based on their emotional experiences, such as enjoyment

and fun during prior interactions with technology (Huang et al., 2024; Lu et al., 2019). In this regard, if a robot is well-designed, able to deliver accurate and efficient care, and possesses effective communication abilities, positive emotional responses like enjoyment, comfort, and trust are likely to be engendered. Other scholars also show that perceived enjoyment acts as a noteworthy precursor of technology adoption while also serving as a mediator between other variables with attitudes and intentions (Holdack et al., 2022). Hence, we propose:

H4. Perceived enjoyment significantly impacts trust (H_{4a}) and the intention to use healthcare service robots (H_{4b}).

3.3. The role of social stimuli in healthcare service robots

The provision of high-quality care via service robots may be considerably influenced by the robots' social attributes. Both the social cues emitted by these robots and their design features hold equal significance. The presence of social support and assistance during interactions with service robots tends to foster a sense of comfort and confidence in the users. Therein lies the perception that interactions with robots mirror human interaction, making them more relatable. This social dimension, encapsulated by perceptions of social interactivity and presence, can significantly direct the acceptance and intention to use service robots.

3.3.1. The impact of perceived social interactivity

Perceived social interactivity captures the essence of situations wherein a user perceives that a robot is responding and behaving in alignment with established social norms. This perceived social interactivity fosters an impression in users that engaging with robots is akin to interacting with fellow humans. In this regard, the more socially capable and interactive a robot is, the more likely it is to kindle interest and foster engagement among users. For such interactions to be effective, robots must possess the ability to execute appropriate actions and display relevant emotions. Notably, the social skills and attractiveness of a robot can serve as compelling motivators for people to interact with service robots (Fernandes & Oliveira, 2021). Echoing this, Kim et al. (2022) provided empirical evidence that social interactivity positively influences both service robots trust and usage intentions. Given the above, we propose:

H5. Perceived social interactivity exerts a significant influence on trust (H_{5a}) and the intention to use healthcare service robots (H_{5b}).

3.3.2. The impact of perceived social presence

Perceived social presence denotes to the degree to which a user considers an entity is authentically "present" (or otherwise), conveying a sense of comfort and togetherness (Lee et al., 2024). This sense of presence can motivate users to interact with virtual entities as though they were real human beings (Kim & Park, 2024). As proposed by Rossi et al. (2020), some individuals perceive robots as social agents capable of exhibiting social behavior. This perceived social presence can impact the acceptance of service robots and, by extension, affect user behavior (Lei & Liu, 2025; Song et al., 2024). In this regard, healthcare service robots with human-like appearances and the ability to interact socially can engage users more effectively. In addition, social presence can directly affect trust, as personal interactions often cultivate trust in an individual (Wirtz et al., 2018). Numerous studies (Felding et al., 2023; Heerink et al., 2009, 2010; Rettinger et al., 2024) on the acceptance and intention to use robots also demonstrate that a stronger sense of social presence results in greater acceptance. Moreover, a strong correlation was found between social presence and robot use inclinations in these studies, wherein a greater sense of social presence corresponded with a higher degree of enjoyment and use propensity of these robots. In this regard, we propose:

H6. Perceived social presence significantly affects trust (H_{6a}) and the intention to use healthcare service robots (H_{6b}).

3.4. The role of trust in healthcare service robots

Trust, a cornerstone of technology acceptance research, is an individual's confidence built on three main pillars: competence, morality, and benevolence (Gefen et al., 2003). This faith allows individuals to feel more secure and confident when engaging with a technology, notably, service robots in healthcare.

Trust's role in encouraging service robots use represents a vital psychological prerequisite for fostering positive human-robot interactions (Miller et al., 2021). Noteworthily, trust is especially critical in uncertain situations: consumers perceive robots as being easier to use and being more useful if the trust they have in these robots is noticeably high (Lee et al., 2018). In this regard, trust mitigates concerns about potential risks and outcomes that may come with utilizing service robots, which may be performing a variety of tasks independently. Trust is also likely to be pivotal for first-time users who may have reservations about the robot's performance and the possible risks tied to their usage, as well as in healthcare settings due to its association with treatment effectiveness and patient satisfaction.

Trust's role as a primary driver of service robot usage intentions has been corroborated by past research (Seo & Lee, 2021). The nexus between trust and technological acceptance has been well-documented across various studies (Meidute-Kavaliauskiene et al., 2021). Tussyadiah et al. (2020) argue that elevated trust levels in service robots are instrumental in shaping positive adoption behaviors. This viewpoint is echoed by research conducted by Park (2020) and Van Pinxteren et al. (2019), which emphasize the pivotality of trust in fostering service robots utilization. Therefore, we propose:

H7. Trust significantly influences the intention to use healthcare service robots and acts as a mediator for perceived ease of use (H_{7a}), perceived usefulness (H_{7b}), perceived anxiety (H_{7c}), perceived enjoyment (H_{7d}), perceived social interactivity (H_{7e}), and perceived social presence (H_{7f}).

3.5. The role of anthropomorphism in healthcare service robots

Understanding the individuals' intention toward service robots necessitates examining the factor of anthropomorphism. Human-like characteristics, including appearance and behaviors, significantly enhance interactions between humans and robots, as evidenced by research (Blut et al., 2021; Lim, Kumar, et al., 2022). Past research indicate a preference for robots that exhibit human-like traits and anthropomorphic features in service interactions (Akdim et al., 2023; Zhang et al., 2022), which are believed to foster user engagement and adoption. Nevertheless, an overabundance of such traits may induce discomfort and adverse reactions in users (Mori, 1970), challenging the notion that robots must closely mimic human characteristics to be deemed socially adept (Breazeal, 2003). Research, including that by Blut et al. (2021), has demonstrated the relationship among anthropomorphism and usage intentions, mediated by a blend of functional, social, and relational factors. Thus, we propose:

H8. Anthropomorphism significantly moderates all relationships from H_1 to H_6.

3.6. The role of subjective norms in healthcare service robots

In healthcare service robotics, subjective norms are critical, acting as societal and interpersonal forces that shape individuals' decisions to embrace or reject these technological innovations; these norms are rooted in the perceived expectations of significant others and the social milieu in which an individual operates (Venkatesh & Davis, 2000). This influence often comes from the social network within which the individual resides. Consequently, the adoption of new technologies,

including healthcare service robots use intentions, can be shaped by these social norms. For instance, patients often solicit opinions from their trusted circles when deciding which doctor to consult or hospital to visit. Similarly, external motivation takes up a significant role in accepting healthcare service robots. This influence is backed by previous studies (Blut et al., 2016; Tavares & Oliveira, 2018), which affirm that social influence affects behavioral intention. To this end, we propose:

H9. Subjective norms significantly moderate all the relationships from H_1 to H_6.

4. Methodology

4.1. Instrumentation

The measurement instruments for our investigation were developed based on existing scholarly work, thus ensuring our questionnaire's robustness. An exhaustive literature review enabled us to source the relevant measures (Table 1). We customized the statements of these items to align more closely with the objective and scope of our research, enabling a more precise measurement of our constructs.

We operationalized functional stimuli, encompassing perceptions of ease of use and usefulness, using a scale adopted from Fernandes and Oliveira (2021). These stimuli, which are inherent in service robots, were represented by four items. The emotional stimuli, including perceptions of anxiety and enjoyment, were assessed using three items each, following the scale by Saari et al. (2022). These emotional aspects gauge patients' emotional perception to interacting with service robots. The social stimuli, namely perceptions of social interactivity and presence, were represented by two items each, examined via scales adapted again from Fernandes and Oliveira (2021). These social dimensions offer insights into the patient's perceived social interactions with the robots. Anthropomorphism and subjective norms, both encompassing two items each, were also adopted from Fernandes and Oliveira (2021). Trust and the intention to use healthcare service robots were represented by four and three items, respectively. These items were derived from scales developed by Mukherjee (2022) for trust and Chuah et al. (2021) for usage intention. These elements evaluate the confidence that patients have in the robots and their intent to use such services in the future.

These measurement items (or questions) were assessable by respondents on a Likert scale that is anchored on a continuum of strongly disagree (1) to strongly agree (5). Likert scales are a well-established method for quantifying attitudes or perceptions, translating subjective opinions into numeric data for statistical analysis (Koo & Yang, 2025). We specifically adopted a five-point format because it strikes an optimal balance between *granularity*—allowing respondents to indicate varying intensities of agreement—and *simplicity*, which reduces cognitive load and survey fatigue. The neutral midpoint also captures truly undecided stances. The questionnaire, which was developed as an online version—deployed online to enable efficient, cost-effective outreach to a geographically dispersed sample and to allow respondents to complete the survey at their own convenience (Lim, 2025)—underwent a *pretest* with two professors who were familiar with healthcare technologies to establish *content validity* and a *pilot study* with 10 potential participants who had experience with service robots in hospitals to establish *face validity*, which enabled us to refine the items for greater clarity and precision before it was administered in the main study (Lim, 2024).

Table 1
Measures and sources.

Measures	Number of items	Source
Functional stimuli		Fernandes and Oliveira (2021)
PEU: Perceived ease of using healthcare service robot	2	
PU: Perceived usefulness of using healthcare service robot	2	
Emotional stimuli		Saari et al. (2022)
PA: Perceived anxiety of using healthcare service robot	3	
PE: Perceived enjoyment of using healthcare service robot	3	
Social stimuli		Fernandes and Oliveira (2021)
PSI: Perceived social interactivity of using healthcare service robot	2	
PSP: Perceived social presence of using healthcare service robot	2	
Moderator		
AN: Anthropomorphism of healthcare service robot	2	Fernandes and Oliveira (2021)
SN: Subjective norms toward healthcare service robot	2	Fernandes and Oliveira (2021)
Outcome		
TR: Trust toward healthcare service robot	4	Mukherjee (2022)
IN: Intention to use healthcare service robots	3	Chuah et al. (2021)

4.2. Sampling

Our research's geographical context was in India, where many hospitals employ service robots to offer a variety of healthcare services, such as cleaning and disinfection, food and medicine delivery, front office assistance, endoscopy and surgery, patient monitoring, and sample collection, among others (Fortune India, 2021; Rife Technologies, 2025).

Following established guidelines for participant recruitment in survey research (Lim, 2025), data were collected using an online survey. We employed a combination of purposive and snowball sampling techniques. Specifically, purposive sampling involved recruiting participants who had prior experience with service robots in hospitals while snowball sampling leveraged these participants' networks to identify and recruit additional eligible participants, thereby ensuring the relevance of our results to the research context. Implied consent was obtained through participants' affirmative indication of previous interaction with hospital service robots, allowing them to proceed to the full questionnaire. All participants were assured anonymity, as no personal information was collected, and participation in the online survey was voluntary with no incentives provided. Participants were informed that there were no right or wrong answers and that they could withdraw at any time before submitting their responses; however, withdrawal after submission was not possible due to anonymity protections. A mix of positive and negatively worded questions were also used in the questionnaire. These procedural steps, in turn, helped to mitigate the potential of common method bias.

Out of the 410 respondents, 12 were excluded due to incomplete or straight-line responses, rendering a total of 398 valid responses for further analysis. This sample size, which exceeds the requirement of 70 responses as per the 10-times rule based on the maximum number of structural paths—in this case, seven paths (ease of use, usefulness, anxiety, enjoyment, social interactivity, social presence, and trust) pointing at the most complex latent endogenous construct (intention) (Lim, 2025)—is sufficient for performing partial least squares structural equation modeling (PLS-SEM) (Hair et al., 2019), which is used to inspect our research model in two stages: measurement model and structural model (Anderson & Gerbing, 1988).

5. Results

5.1. Profile of participants

A demographic overview of participants, including characteristics such as gender, age, and educational attainment, which are pertinent to understanding their responses, is provided in Table 2. The majority of participants identified as male (51.3%), with females constituting the

Table 2
Profile of participants.

Characteristic		*n* (398)	% (100)
Gender	Female	194	48.7
	Male	204	51.3
Age (in years)	18–25 years	108	27.1
	26–35 years	146	36.7
	36–45 years	132	33.1
	46–55 years	6	1.5
	56–65 years	5	1.3
	>65 years	1	0.3
Education	High school or lower	11	2.8
	Diploma	7	1.8
	Undergraduate bachelor's degree	188	47.2
	Postgraduate certificate or master's degree	172	43.2
	Postgraduate doctoral degree	20	5.0

remaining 48.7%. The predominant age group was 26–35 years old (36.7%), followed by individuals aged 36 to 45 (33.1%), and 18 to 25 (27.1%). Participants older than 46 years were scarce. As for educational attainment, nearly half of the participants held an undergraduate degree (47.2%) while 43.2% possessed a postgraduate degree. A smaller proportion had completed doctoral programs, high school, or diploma courses.

5.2. Measurement model and common method bias evaluation

In our analysis, detailed in Table 3, we scrutinized the measurement model for both reliability and validity, adhering to robust statistical standards. The analysis revealed that all items (apart from TR2, which was retained on the basis of substantive significance or theoretical relevance) significantly loaded on their respective factors with values above the 0.70 benchmark, as suggested by Hair et al. (2019), indicating strong factor loadings.

To assess the internal consistency of the constructs, we employed both Cronbach's alpha (CA) and composite reliability (CR) metrics. The CA values spanned from 0.703 to 0.889, exceeding the established threshold of 0.70 (Nunnally, 1978), while CR values similarly surpassed the 0.70 benchmark (Hair et al., 2019), affirming the *reliability* of our constructs.

Through confirmatory factor analysis, we also examined the constructs for convergent and discriminant validity (Lim, 2024). The constructs had adequate *convergent validity*, with average variance extracted (AVE) values surpassing the 0.50 minimum requirement (Hair et al., 2019). The Fornell and Larcker (1981) criterion further validated the

Table 3
Measurement items statistics.

Construct and item		Convergent validity		Internal consistency or reliability	
		Factor loading	Average variance extracted	Cronbach's alpha	Composite reliability
Functional stimuli					
PEU: Perceived ease of using healthcare service robot			0.837	0.812	0.911
PEU1	Healthcare service robots are easy to use.	0.880			
PEU2	Healthcare service robots are self-paced and can be used independently.	0.948			
PU: Perceived usefulness of using healthcare service robot			0.771	0.808	0.871
PU1	Healthcare service robots offer useful services.	0.844			
PU2	Healthcare service robots offer significant assistance.	0.911			
Emotional stimuli					
PA: Perceived anxiety of using healthcare service robot			0.647	0.703	0.839
PA1	Interacting with healthcare service robots makes me feel nervous at times.*	0.725			
PA2	Healthcare service robots occasionally make me feel uncomfortable.*	0.919			
PA3	Healthcare service robots sometimes cause feelings of uneasiness.*	0.905			
PE: Perceived enjoyment of using healthcare service robot			0.791	0.870	0.919
PE1	Using healthcare service robots can be an enjoyable experience.	0.886			
PE2	The process of using a healthcare service robot is pleasant.	0.877			
PE3	Interacting with healthcare service robots can be a fun experience.	0.904			
Social stimuli					
PSI: Perceived social interactivity of using healthcare service robot			0.757	0.889	0.861
PSI1	Healthcare service robots interact in a pleasing manner.	0.815			
PSI2	Healthcare service robots understand my healthcare needs.	0.922			
PSP: Perceived social presence of using healthcare service robot			0.748	0.825	0.856
PSP1	I experience real-time interactions with healthcare service robots.	0.841			
PSP2	I feel the social presence of healthcare service robots.	0.888			
Moderator					
AN: Anthropomorphism of healthcare service robot			0.839	0.809	0.913
AN1	Healthcare service robots offer services similar to those provided by humans.	0.907			
AN2	Sometimes, I perceive healthcare service robots to be almost human.	0.925			
SN: Subjective norms toward healthcare service robot			0.606	0.851	0.855
SN1	I prefer to use healthcare service robot services based on others' recommendations.	0.763			
SN2	I use healthcare service robots based on the recommendations of friends and relatives.	0.794			
Outcome					
TR: Trust toward healthcare service robot			0.538	0.718	0.821
TR1	Healthcare service robots provide dependable services.	0.769			
TR2	I place a high level of confidence in healthcare service robots.	0.601			
TR3	Healthcare service robots are highly reliable and do not often make mistakes or errors.	0.848			
TR4	Healthcare service robots can be trusted to handle the personal data of patients.	0.792			
IN: Intention to use healthcare service robots			0.805	0.884	0.925
IN1	I am eager to avail myself to the services offered by healthcare service robots.	0.911			
IN2	I am willing to use healthcare service robots in the future.	0.883			
IN3	I am in favor of using healthcare service robots in the future.	0.898			

Notes: * = Negatively worded and recoded.

discriminant validity of the constructs, showing that square root of AVE values (diagonal entries in Table 4) were more than inter-construct correlations, indicating distinctiveness among constructs. Moreover, we applied the heterotrait-monotrait (HTMT) ratio as an additional test for discriminant validity. The HTMT values, presented in Table 5, were all below the 0.85 threshold (Henseler et al., 2015), confirming the constructs' *discriminant validity*.

We also examined potential *multicollinearity* issues among our constructs. Variance inflation factor (VIF) values in the outer model ranged from 1.047 to 2.978, remaining well below the critical value of 5.0 (Hair et al., 2014), thus negating concerns of multicollinearity. To address the possibility of *common method bias*, we ensured that the VIF values did not exceed the 3.30 limit as per Kock (2017), ensuring the model's robustness against such bias.

Lastly, the model's fit was examined via the bootstrapped standardized root mean square residual (SRMR) value in the SmartPLS software, which was found to be 0.064, below the maximum threshold of 0.08, indicating a satisfactory fit of the measurement model to the data.

5.3. Structural model evaluation and hypotheses testing

5.3.1. Direct effects

As depicted in Fig. 2 and Table 6, the direct effects results supported all hypotheses, except for those involving perceived usefulness, with a 95 % confidence level. To begin, functional elements such as perceived ease of use exhibited a significantly positive effect on trust ($\beta = 0.079$, $p < 0.01$) and the intention to use service robots ($\beta = 0.073$, $p < 0.05$), thereby corroborating H_{1a} and H_{1b}. Conversely, perceived usefulness had no significant effect on trust ($\beta = 0.019$, $p = 0.611$) or the intention to use service robots ($\beta = 0.048$, $p = 0.413$), which resulted in H_{2a} and H_{2b} not being supported. Furthermore, emotional elements such as perceived anxiety and enjoyment demonstrate significantly positive effects on both trust and the intention to use healthcare service robots. This finding provided support for H_{3a} ($\beta = 0.148$, $p < 0.001$), H_{3b} ($\beta = 0.067$, $p < 0.05$), H_{4a} ($\beta = 0.228$, $p < 0.001$), and H_{4b} ($\beta = 0.074$, $p < 0.05$). Moreover, social factors like perceived social interactivity and presence also had a substantial impact on trust and the intention to use service robots, thereby supporting H_{5a} ($\beta = 0.093$, $p < 0.01$), H_{5b} ($\beta = 0.112$, $p < 0.01$), H_{6a} ($\beta = 0.092$, $p < 0.01$), and H_{6b} ($\beta = 0.117$, $p < 0.001$). Finally, trust's direct effect on the intention to use healthcare service robots was also substantial ($\beta = 0.304$, $p < 0.001$), rendering support for H_7.

5.3.2. Mediation effects

Trust is considered crucial when embracing novel technology. Hence, we examined trust's mediating effect between the relationship of functional, emotional, and social elements and the intention to use healthcare service robots. Findings from the research indicate that trust significantly mediated the relationship between all elements except perceived usefulness and the intention to use service robots ($\beta = 0.006$, $p = 0.613$). This mediation was significant for perceived ease of use ($\beta = 0.024$, $p < 0.05$), robot-induced anxiety ($\beta = 0.045$, $p < 0.001$), enjoyment ($\beta = 0.069$, $p < 0.001$), social interactivity ($\beta = 0.028$, $p < 0.01$), and social presence ($\beta = 0.028$, $p < 0.01$). Thus, H_{7a}, H_{7c}, H_{7d}, H_{7e}, and H_{7f} are supported, but not H_{7b}.

5.3.3. Moderation effects

We further performed a moderation analysis to probe the moderating influence of anthropomorphism and subjective norms on the relationships between antecedents and outcomes. The findings, presented in Table 6, revealed subjective norms' significant moderating effect on the relationship between perceived social interactivity and the intention to use healthcare service robots ($\beta = 0.124$, $p < 0.05$). No significant moderating effects were found in any other relationships. Thus, only H_{9j} is supported, with the rest of H_8 and H_9 not supported.

6. Discussion

The advancement of automation technologies has notably enhanced the presence and potential of service robots in the healthcare sector. Despite this progression, the exploration into how these robots are embraced and integrated into healthcare practices is still in its nascent stages. It is essential to dissect the specific beliefs and motivations that encourage users to welcome service robots into healthcare settings. Attributes such as the human-like appearance of robots (Lim, Kumar, et al., 2022; Xin & Liu, 2025) and the nature of human-robot interactions (de Oliveira Santini et al., 2025; Wang et al., 2025) are pivotal in forming users' perceptions and evaluations of these technologies. Therefore, our research focuses on uncovering the key factors that influence the readiness to adopt service robots in healthcare environments. We corroborate the reliability and validity of our proposed service robot acceptance model (sRAM) through a comprehensive empirical investigation, gathering insights from 398 individuals who have previously interacted with service robots in hospitals.

In technology adoption, users typically prioritize functional attributes like ease of use when evaluating new systems (Davis, 1989; Lim, 2018). For healthcare service robots, a user-friendly interface and straightforward functionality are crucial for increasing acceptance (Costanzo et al., 2024). Consistent with prior research, our findings reinforce the notion that the likelihood of patients engaging with technology-based services rises when the technology is perceived as a straightforward, user-friendly interface with easy navigation (Mechta et al., 2024). Our research further supports the significant role of ease of use in building trust and fostering the intention to adopt service robots in healthcare settings, aligning with earlier evidence that links low effort

Table 4
Correlation and square-root of average variance extracted (bold diagonal).

Construct	AN	IN	PA	PE	PEU	PSI	PSP	PU	SN	TR
AN	**0.916**									
IN	−0.037	**0.897**								
PA	0.527	0.016	**0.804**							
PE	0.361	0.031	0.799	**0.889**						
PEU	0.389	−0.088	0.345	0.280	**0.915**					
PSI	0.752	−0.016	0.423	0.287	0.364	**0.870**				
PSP	0.519	0.089	0.471	0.288	0.320	0.560	**0.865**			
PU	0.389	−0.050	0.375	0.314	0.809	0.366	0.315	**0.878**		
SN	0.616	−0.033	0.468	0.341	0.636	0.613	0.482	0.646	**0.779**	
TR	0.326	0.040	0.310	0.296	0.222	0.241	0.197	0.198	0.242	**0.733**

Notes: AN: Anthropomorphism of healthcare service robot. IN: Intention to use healthcare service robots. PA: Perceived anxiety of using healthcare service robot. PE: Perceived enjoyment of using healthcare service robot. PEU: Perceived ease of using healthcare service robot. PSI: Perceived social interactivity of using healthcare service robot. PSP: Perceived social presence of using healthcare service robot. PU: Perceived usefulness of using healthcare service robot. SN: Subjective norms toward healthcare service robot. TR: Trust toward healthcare service robot.

Table 5
Heterotrait-monotrait (HTMT) ratio of correlation.

Construct	AN	IN	PA	PE	PEU	PSI	PSP	PU	SN
AN									
IN	0.051								
PA	0.742	0.035							
PE	0.426	0.035	0.005						
PEU	0.482	0.088	0.512	0.341					
PSI	0.025	0.065	0.668	0.373	0.512				
PSP	0.71	0.117	0.794	0.376	0.45	0.736			
PU	0.518	0.061	0.575	0.409	0.08	0.556	0.47		
SN	0.149	0.07	0.035	0.616	0.23	0.251	0.699	0.324	
TR	0.401	0.058	0.415	0.362	0.271	0.302	0.274	0.253	0.443

Notes: AN: Anthropomorphism of healthcare service robot. IN: Intention to use healthcare service robots. PA: Perceived anxiety of using healthcare service robot. PE: Perceived enjoyment of using healthcare service robot. PEU: Perceived ease of using healthcare service robot. PSI: Perceived social interactivity of using healthcare service robot. PSP: Perceived social presence of using healthcare service robot. PU: Perceived usefulness of using healthcare service robot. SN: Subjective norms toward healthcare service robot. TR: Trust toward healthcare service robot.

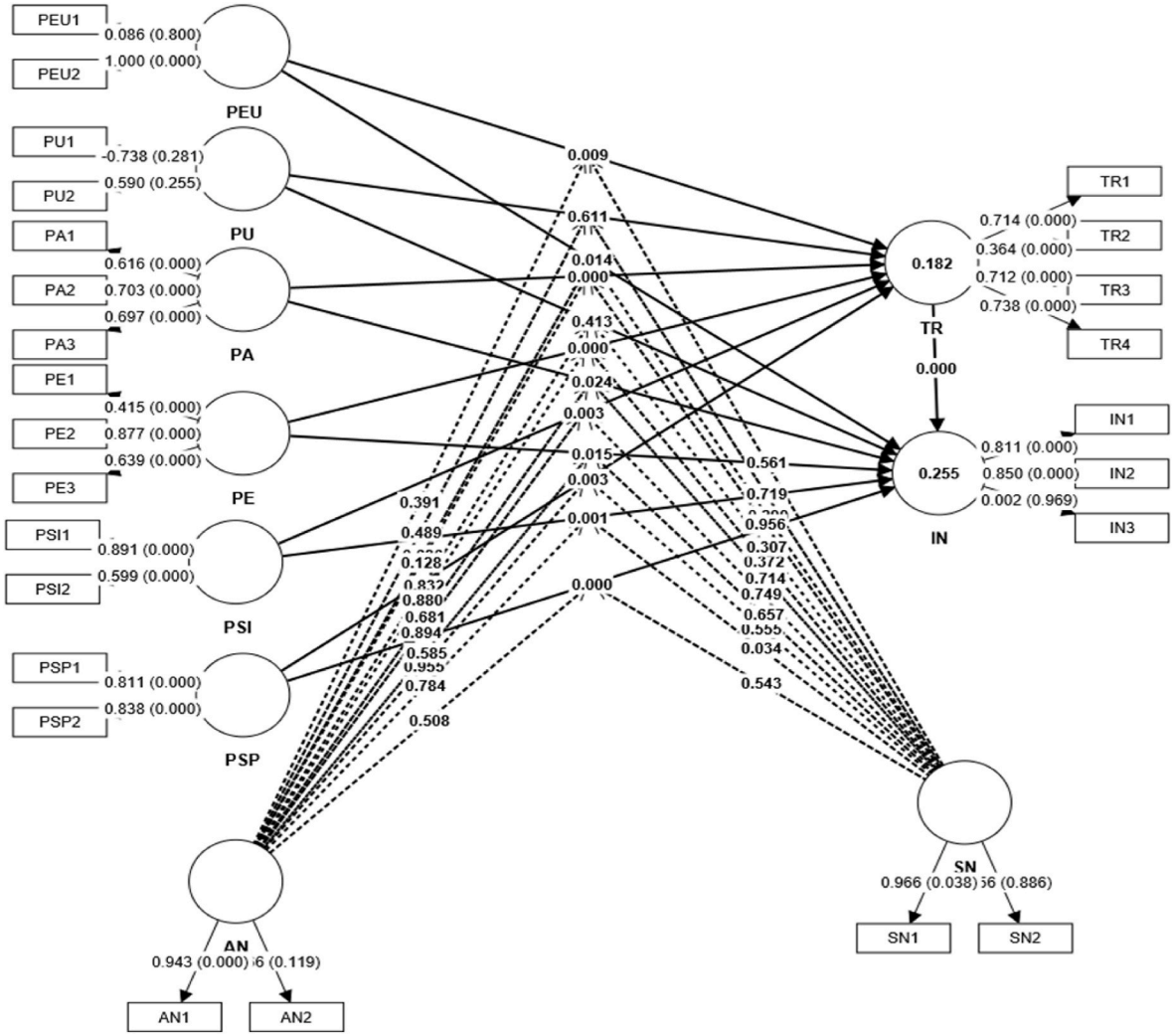


Fig. 2. Structural model
Notes: AN: Anthropomorphism of healthcare service robot. IN: Intention to use healthcare service robots. PA: Perceived anxiety of using healthcare service robot. PE: Perceived enjoyment of using healthcare service robot. PEU: Perceived ease of using healthcare service robot. PSI: Perceived social interactivity of using healthcare service robot. PSP: Perceived social presence of using healthcare service robot. PU: Perceived usefulness of using healthcare service robot. SN: Subjective norms toward healthcare service robot. TR: Trust toward healthcare service robot.

expectancy with the willingness to adopt new technologies and extending the generalizability of this evidence from healthcare professionals to patients (Kim, Choi, & Fotso, 2024). Nonetheless, these confirmations underscore a critical implication: healthcare providers must recognize ease of use not just as a basic requirement but as a strategic imperative, prompting continuous innovation in design that reduces cognitive load and enhances intuitive interactions. Indeed, recent empirical evidence underscores that patients frequently avoid or

Table 6
Structural model results.

Hypothesis	Relationship	β	*t*-statistic	*p*-value	Outcome
Direct effect					
H_{1a}	Perceived ease of use → Trust	0.079	2.620	0.009**	Supported
H_{1b}	Perceived ease of use → Intention to use	0.073	2.468	0.014*	Supported
H_{2a}	Perceived usefulness → Trust	0.019	0.508	0.611	Not supported
H_{2b}	Perceived usefulness → Intention to use	0.048	0.818	0.413	Not supported
H_{3a}	Perceived anxiety → Trust	0.148	5.069	0.000***	Supported
H_{3b}	Perceived anxiety → Intention to use	0.067	2.253	0.024*	Supported
H_{4a}	Perceived enjoyment → Trust	0.228	7.582	0.000***	Supported
H_{4b}	Perceived enjoyment → Intention to use	0.074	2.434	0.015*	Supported
H_{5a}	Perceived social interactivity → Trust	0.093	2.937	0.003**	Supported
H_{5b}	Perceived social interactivity → Intention to use	0.112	3.316	0.001**	Supported
H_{6a}	Perceived social presence → Trust	0.092	2.946	0.003**	Supported
H_{6b}	Perceived social presence → Intention to use	0.117	3.895	0.000***	Supported
H_{7}	Trust → Intention to use	0.304	10.839	0.000***	Supported
Mediation effect					
H_{7a}	Perceived ease of use → Trust → Intention to use	0.024	2.559	0.011*	Supported
H_{7b}	Perceived usefulness → Trust → Intention to use	0.006	0.505	0.613	Not Supported
H_{7c}	Perceived anxiety → Trust → Intention to use	0.045	4.546	0.000***	Supported
H_{7d}	Perceived enjoyment → Trust → Intention to use	0.069	5.927	0.000***	Supported
H_{7e}	Perceived social interactivity → Trust → Intention to use	0.028	2.783	0.005**	Supported
H_{7f}	Perceived social presence → Trust → Intention to use	0.028	2.835	0.005**	Supported
Moderation effect					
H_{8a}	Anthropomorphism × Perceived ease of use → Trust	−0.005	0.151	0.880	Not supported
H_{8b}	Anthropomorphism × Perceived ease of use → Intention to use	−0.016	0.547	0.585	Not supported
H_{8c}	Anthropomorphism × Perceived usefulness → Trust	0.023	0.691	0.489	Not supported
H_{8d}	Anthropomorphism × Perceived usefulness → Intention to use	−0.006	0.213	0.832	Not supported
H_{8e}	Anthropomorphism × Perceived anxiety → Trust	0.049	1.522	0.128	Not supported
H_{8f}	Anthropomorphism × Perceived anxiety → Intention to use	0.016	0.411	0.681	Not supported
H_{8g}	Anthropomorphism × Perceived enjoyment → Trust	−0.023	0.858	0.391	Not supported
H_{8h}	Anthropomorphism × Perceived enjoyment → Intention to use	0.014	0.473	0.636	Not supported
H_{8i}	Anthropomorphism × Perceived social interactivity → Trust	−0.002	0.056	0.955	Not supported
H_{8j}	Anthropomorphism × Perceived social interactivity → Intention to use	0.028	0.662	0.508	Not supported
H_{8k}	Anthropomorphism × Perceived social presence → Trust	0.004	0.133	0.894	Not supported
H_{8l}	Anthropomorphism × Perceived social presence → Intention to use	−0.01	0.274	0.784	Not supported
H_{9a}	Subjective norms × Perceived ease of use → Trust	0.019	0.582	0.561	Not supported
H_{9b}	Subjective norms × Perceived ease of use → Intention to use	−0.031	1.059	0.290	Not supported
H_{9c}	Subjective norms × Perceived usefulness → Trust	−0.05	1.022	0.307	Not supported
H_{9d}	Subjective norms × Perceived usefulness → Intention to use	−0.013	0.36	0.719	Not supported
H_{9e}	Subjective norms × Perceived anxiety → Trust	−0.002	0.056	0.956	Not supported
H_{9f}	Subjective norms × Perceived anxiety → Intention to use	−0.011	0.366	0.714	Not supported
H_{9g}	Subjective norms × Perceived enjoyment → Trust	0.033	0.892	0.372	Not supported
H_{9h}	Subjective norms × Perceived enjoyment → Intention to use	0.013	0.444	0.657	Not supported
H_{9i}	Subjective norms × Perceived social interactivity → Trust	0.009	0.319	0.749	Not supported
H_{9j}	Subjective norms × Perceived social interactivity → Intention to use	0.124	2.116	0.034*	Supported
H_{9k}	Subjective norms × Perceived social presence → Trust	0.02	0.590	0.555	Not supported
H_{9l}	Subjective norms × Perceived social presence → Intention to use	−0.02	0.609	0.543	Not supported

Notes: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

discontinue the use of healthcare technologies perceived as difficult or non-intuitive, including service robots (El-Gazar et al., 2024). Therefore, optimizing ease of use becomes essential for healthcare providers seeking to enhance patients' trust and strengthen their intention to embrace technological solutions such as service robots.

However, contrary to TAM and other studies that highlight perceived usefulness as a pivotal factor in the acceptance and intention to use technology, our research reveals a divergent finding. Perceived usefulness did not significantly influence the trust in or the intention to adopt healthcare service robots in our research. This deviation challenges the established notion supported by prior research that perceived usefulness is a strong predictor of behavioral intention (Gani et al., 2024). This observation is critical and novel, as it suggests that, despite the prevailing assumption in technology adoption (Lim, 2018), usefulness alone may not suffice to drive adoption in healthcare, where human interaction and emotional reassurance remain paramount. This distinction—when considered in tandem with the other insights herein—implies that healthcare technologies, especially service robots, are judged less by their inherent functional capabilities and more by their ease of use, emotional compatibility, and social acceptability. Consequently, healthcare providers should reconsider their strategic use of service robots, refraining from delegating value-creating activities entirely to these technologies. Instead, healthcare providers should deploy service robots primarily as facilitators, complementing human care rather than substituting it, thus aligning robot adoption strategies with patient expectations and preferences in healthcare delivery.

Emotional reactions from patients also play a noteworthy role in shaping their responses toward healthcare service robots. Interactions with robots can elicit various emotional responses (Filieri et al., 2022). In a hospital setting, some patients may experience anxiety, fear, or sadness due to their health conditions, discomfort, or pain. Conversely, others may feel positive emotions, such as relief from illness or gratitude toward healthcare providers. Patients' emotional states during interactions with robots can therefore vary significantly, depending on previous interactions with robots, their attitudes toward technology, and their overall mental and emotional states. Some patients may resist adopting robots due to feelings of discomfort (Tsai et al., 2020) while others might appreciate the convenience and efficiency of robots in hospital settings (Wang et al., 2024). Our research specifically finds that reducing robot-induced anxiety and enhancing enjoyment significantly strengthen patients' trust and intention to adopt healthcare service robots. This insight corroborates with the work of Li et al. (2024), who suggested that perceived enjoyment during interaction with service robots alleviates feelings of anxiety and leads to acceptance. Past scholars

also found that users with no experience using digital technology are likely to feel anxious when they interact with humanoid robots (Chattaraman et al., 2019). Other studies have confirmed that perceived enjoyment positively impacts service robot acceptance within the domestic or home environment (Turja et al., 2020). Critically, these findings imply that healthcare providers should proactively design robot interactions to minimize anxiety and maximize positive emotional experiences—transforming emotional management from a peripheral consideration into a central strategy for fostering patient trust and acceptance of robotic healthcare services.

Social dimensions, such as the sense of interaction and presence offered by robots, foster their acceptance (Wirtz et al., 2018). Our research underscores a notable link between the levels of social interactivity and presence perceived in healthcare robots, trust in these technologies, and the subsequent willingness to utilize them. This finding is in harmony with research (Fernandes & Oliveira, 2021; Shi et al., 2024) that underscores the importance of these social factors in the acceptance and integration of service robots in healthcare settings. Critically, these insights highlight that healthcare providers must move beyond designing robots solely as functional tools. Instead, healthcare providers should strategically prioritize developing robots that interact in pleasing ways, demonstrate an understanding of individual healthcare needs, and facilitate interactions perceived by patients as socially present and occurring in real-time. This approach transforms patient–robot engagements from purely transactional interactions into meaningful social connections, significantly strengthening patient trust and their intention to adopt healthcare service robots.

Trust also emerges as a key determinant in the decision to adopt service robots. Individuals exhibiting significant trust and engagement with these robots are often more inclined to find the interaction satisfying, which, in turn, catalyzes their intention to continue using the technology. This direct linkage between trust and the intent to use is supported by findings from Kraus et al. (2024), Song et al. (2022), and Van Pinxteren et al. (2019), who suggest that trust in human-robot interactions significantly propels the behavioral intention toward technology use. Moreover, our research indicates that trust acts as a critical mediator in the relationship between the perceived functional, emotional, and social qualities of service robots and their acceptance, with the exception of perceived usefulness. Critically, these findings extend previous research by indicating that trust-building strategies must simultaneously leverage multiple dimensions of robot interactions—functional ease (ease of use), emotional reassurance (low anxiety, high enjoyment), and social connectedness (interactivity, sense of presence)—to effectively enhance acceptance. Rather than simply replicating earlier calls to improve trust, our results imply a more integrative approach, where healthcare providers strategically align robot design and communication to holistically foster patient trust and drive intentions to adopt healthcare service robots.

Finally, the role of anthropomorphism and subjective norms as moderators between the antecedent elements and trust, and intention to use, appears largely insignificant. This finding suggests that in healthcare settings, patients' perceptions and intentions regarding service robots may be driven primarily by their direct interactions and immediate experiences—such as ease of use, anxiety reduction, enjoyment, and social interactions—rather than by the degree of human-like features in the robots or external social influences. Given that patients typically prioritize their own healthcare needs, comfort, and safety over aesthetic elements or external influences, anthropomorphism and subjective norms may hold less relevance in influencing trust and adoption intentions within healthcare. Consequently, healthcare providers should strategically emphasize and enhance the direct experiential aspects identified as significant in this research, such as ease of use, positive emotional engagement, and meaningful social interactions, rather than relying on anthropomorphic design or social conformity as primary drivers for service robot acceptance.

7. Conclusion

7.1. Theoretical contributions

Our research delivers notable theoretical contributions in the rapidly evolving field of service robots in the healthcare sector. A key focus of our research is the application and expansion of the sRAM within a healthcare context.

Firstly, we offer a comprehensive approach to the model by incorporating constructs like functional, emotional, and social elements related to service robots. Each of these constructs was empirically validated to evaluate their impact on service robots trust and usage intentions in a healthcare setting. The multifaceted approach herein provides a more holistic encapsulation of the factors shaping service robot acceptance, responding to the issue highlighted in our problem statement about the fragmented nature of current models (e.g., Fuentes-Moraleda et al., 2020; Kao & Huang, 2023).

Secondly, we provide a contribution to the relatively sparse body of research regarding emotional evaluations of using service robots in healthcare settings (Ahuja, 2024; Shank et al., 2019; Tsai et al., 2021). Recognizing that healthcare providers, caregivers, and patients may have varying emotional responses to service robots depending on the context, we expanded the sRAM by introducing emotional elements such as robot-induced perceived anxiety and perceived enjoyment. These constructs are posited as determinants that could significantly influence service robots trust and use. This novel contribution presents a better understanding of the emotional elements related to healthcare robot usage, a critical aspect often overlooked in current models.

Thirdly, we leverage the S-O-R model to shed light on the dynamics between functional, emotional, and social factors, trust, and the intention to engage with service robots. In our analysis, variables such as perceived ease of use, anxiety, enjoyment, social interactivity, and social presence are considered stimuli that, through the lens of trust (the organism), influence the consequential response: the intention to adopt service robots in healthcare settings. The exploration of trust's mediating role reveals that when combined with these antecedents, trust significantly enhances the likelihood of service robot adoption in healthcare settings. This finding underscores the value of integrating the S-O-R model into our investigation, distinguishing our work by elucidating the complex interplay of these elements.

Finally, we scrutinize the moderating role of anthropomorphism and subjective norms on the interrelationships between antecedents and outcomes of healthcare service robot acceptance. This analysis further enriches our understanding of how social elements shape service robot acceptance, notably by demonstrating that subjective norms significantly moderate the influence of perceived social interactivity on usage intentions within healthcare. This finding implies that patients' intentions to use healthcare service robots are highly sensitive to social expectations, especially when these robots involve social interaction. In other words, patients are more likely influenced by the opinions of important others when using the service robot involves visible social interactions endorsed or positively perceived by their social circles. Conversely, the lack of significant moderation for other antecedents indicates that patients do not rely heavily on social cues when evaluating functional or emotional aspects of robots, reinforcing the idea that healthcare providers should strategically leverage social influence (e.g., highlighting that healthcare professionals and patients endorse and interact with robots) mainly in scenarios emphasizing interpersonal interactions, rather than applying it uniformly across all robot attributes.

This research therefore offers a comprehensive model for understanding the acceptance of service robots in healthcare. This model not only unifies fragmented elements within current theories but also introduces novel constructs and relationships, thereby providing a solid foundation for future research.

7.2. Managerial implications

The findings from our research underscore that patients' willingness to embrace healthcare service robots hinges on functional, emotional, and social cues, which jointly influence their trust in, and intention to use, these technologies. Healthcare providers should thus concentrate their efforts on strategies that encourage intuitive robot interactions, foster positive emotional experiences, and leverage social stimuli judiciously—rather than relying solely on the robots' perceived usefulness or anthropomorphic design elements.

To begin, healthcare providers must ensure that the robots are as intuitive and self-paced as possible, given that perceived ease of use emerged as a significant driver of both trust and adoption intention. This entails equipping robots with clear on-screen instructions, easily navigable user interfaces, and optional tutorial modes that allow patients to learn and operate at their own speed. Such design features are crucial for minimizing patients' confusion and hesitancy, and, in turn, strengthening the acceptability of service robots in healthcare.

At the same time, healthcare providers should pay close attention to emotional factors. Reducing patients' anxiety is critical, not only for bolstering trust but also for increasing patients' desire to engage with service robots. This suggests that communication and design strategies must actively alleviate discomfort. It may be helpful to avoid intimidating aesthetics, ensure transparent and reassuring messaging about how robots operate, and provide avenues for patients to get quick answers to their concerns. Similarly, fostering perceived enjoyment is essential. Simple additions—like friendly on-screen greetings, positive feedback messages, or light personalization—should enhance patient enjoyment during interactions, thereby creating a more welcoming environment.

Furthermore, the social features of service robots—reflected in perceived social interactivity and perceived social presence—significantly influence both trust and usage intentions. Healthcare providers can emphasize these interactive capabilities by ensuring that service robots communicate in real time with patients, understand basic patient needs, and respond in ways that convey empathy or attentiveness. Integrating context-awareness and voice recognition can enable service robots adapt to each patient's unique situation. These enhancements, in turn, elevate the patient experience from a transactional encounter to a meaningful interaction, a shift that substantially increases trust in robotic services.

In addition, subjective norms proved to be a potent moderator for social interactivity but showed less relevance for purely functional or emotional factors. Thus, healthcare providers could strategically showcase professional or peer endorsements specifically in contexts where service robots carry out socially engaging tasks. Publicly highlighting endorsements from well-respected clinicians, or encouraging patients to share positive experiences with service robots, leverages the power of social approval. However, this approach needs to be targeted at moments of visible social interaction, rather than uniformly applied across all robot functions.

Moreover, trust remains the central mechanism driving a patient's intention to adopt healthcare service robots. Consequently, healthcare providers should mitigate breakdowns and failures, where possible, by implementing maintenance protocols and ensure healthcare staff are well-trained to handle technical glitches swiftly. Equally important is communicating clear safeguards for data privacy, such as encryption and regulated data handling, which reassure patients that their personal health information is protected.

However, our results show that perceived usefulness neither predicts trust nor intention to adopt, indicating that adding advanced or extra functionalities does not necessarily sway patients. Since perceived usefulness was not a significant predictor, healthcare providers should not over-invest in showcasing a broad array of sophisticated features. Instead, the priority should be to deliver core assistance functions reliably, enabling patients to recognize a basic level of support without feeling overwhelmed by excessive, possibly unnecessary, technological capabilities.

Similarly, efforts to make service robots appear more human-like do not appear to be a decisive factor in healthcare, as anthropomorphism did not significantly moderate relationships between antecedents and outcomes. Patients seem to care less about human-like features and more about practical, emotional, and social benefits. Consequently, healthcare providers should prioritize investments in functionalities that meaningfully enhance patient experiences—for example, simpler interfaces and improved interactive responses—rather than focusing on physical similarities to humans.

These insights collectively highlight an integrated approach for healthcare providers to successfully implement service robots, wherein centering on simplicity (ease of use), emotional well-being (low anxiety, high enjoyment), social engagement (interactivity, presence), and trust-building measures (data security, reliability, staff preparedness) can enable healthcare providers to nurture patient trust and encourage their acceptance of service robots. This evidence-based strategy will help solidify service robots' role as effective facilitators of care, aligned with patient expectations and conducive to the tech evolution of healthcare delivery.

7.3. Limitations and future directions

Despite the contributions of this research, several limitations must be acknowledged, which, in turn, pave the way for future research. To put it simply, this research is exploratory in nature and relies on self-reported responses obtained via online surveys from patients who visited hospitals equipped with service robots in India. Therefore, future research could pursue a confirmatory route using field data through experiments (e.g., testing various ways to enhance ease of use, reduce anxiety, as well as improve enjoyment, social interactivity, and social presence to determine which approach is most effective among patients with varying backgrounds and conditions) and longitudinal follow-ups conducted in person at diverse healthcare settings (e.g., clinics, hospitals, nursing homes), employing healthcare technologies that may include but also extend beyond service robots across urban, semi-urban/semi-rural, and rural areas in other developed and developing countries.

CRediT authorship contribution statement

Weng Marc Lim: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **K. Mohamed Jasim:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **A. Malathi:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

Data can be made available upon reasonable request.

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Weng Marc Lim is a Distinguished Professor and the Dean of Sunway Business School at Sunway University, an ASU-Cintana Alliance Global Partner Affiliate Faculty at Arizona State University, and an Adjunct Professor at Swinburne University of Technology and Sungkyunkwan University. He has served as the Editor in Chief of *Global Business and Organizational Excellence*; Deputy Editor of *Journal of Global Marketing*; Associate Editor of *Journal of Business Research*, *Journal of Consumer Behaviour*, and *Journal of Strategic Marketing*; and Regional Editor of *The Service Industries Journal*. He has authored more than 100 manuscripts in journals ranked 'A*' and 'A' such as *Australasian Marketing Journal*, *European Journal of Marketing*, *Industrial Marketing Management*, *Journal of Business Research*, *Journal of Business and Industrial Marketing*, *Journal of Consumer Behaviour*, *Journal of Consumer Marketing*, *International Journal of Consumer Studies*, *Journal of Brand Management*, *Journal of Product and Brand Management*, *Journal of Retailing and Consumer Services*, *Journal of International Marketing*, *Journal of Marketing Management*, *Journal of Strategic Marketing*, *Marketing Theory*, *Marketing Intelligence & Planning*, and *Psychology & Marketing*, among others. He has also led high-level discussions at the *AppliedHE*, *Association to Advance Collegiate Schools of Business* (*AACSB*), *Association of MBAs* (*AMBA*), *Business Graduates Association* (*BGA*), *Quacquarelli Symonds* (*QS*), *Times Higher Education* (*THE*), *United Nations Educational, Scientific and Cultural Organization* (*UNESCO*) and *Principles of Responsible Management Education* (*PRME*), and *World Economic Forum* (*WEF*). Contact: @limwengmarc on Instagram and Twitter (X), LinkedIn, or his personal homepage at https://www.wengmarc.com.

K. Mohamed Jasim is an Assistant Professor of VIT Business School at Vellore Institute of Technology in India. He can be contacted at mohamedjasim.k@vit.ac.in.

A. Malathi is a Research Scholar of VIT Business School at Vellore Institute of Technology in India. She can be contacted at malathi.a@vit.ac.in.